Contents

1 Open and closed sets

Definition. An interval on the number line is called open if it goes from A to B without containing the endpoints A and B, i.e. it does not contain its boundary.

Definition. $\mathbb{R}$ is the set of real numbers and $\mathbb{R}^n$ is the set corresponding to n-dimensional space

The same can be said for subsets of the plane or higher dimensional spaces – they are open if they do not contain any of their boundary. They are closed if they contain all of their boundary. The formal definition of a boundary is a point where no matter how small of a circle or ball you make around that point, it will contain points both inside and outside the set. Note that since an open set contains none of its boundary, it has the property that for any point in the open set, all points sufficiently close to that point are in that set, so open sets must always have some thickness to them.

This is related to the idea that there is no smallest real number greater than another real number, and whether sets are open or closed turns out to be surprisingly important. Also, arbitrary unions of open sets are open since each point has a "ball" around it inside one of the open sets in the union and thus the whole thing, similarly arbitrary intersections of closed sets are closed. The reverse implications only hold for finite unions and intersections (since infinitely many balls intersection could be a point but finitely many is a ball). Sets can be both open and closed (such as $\mathbb{R}$), and sets can be neither open nor closed.

A useful way to think of closed sets is as sets where the set of everything not in the closed set forms an open set.

We usually call open intervals (a,b) meaning everything from a to b not inclusive, and closed intervals [a,b] to mean everything from a to b inclusive, and [a,b) means everything from a to b including a but not b.

Lemma. Suppose we have a sequence of closed and bounded sets $C_n \subseteq \mathbb{R}^k$ such that for all n we have $C_n \supseteq C_{n+1}$ and they are all non-empty, then their intersection is non-empty.

Proof. As we discuss later in this document, the Bolzano-Weierstrass theorem holds in $\mathbb{R}^k$ - bounded sequences have convergent subsequences. We will pick any sequence (a_n) such that $a_n \in C_n$, then this sequence is bounded as it is contained in C_1 so it has a convergent subsequence, and its limit is in C_j for every j since its elements are eventually all in C_j and C_j is closed.

□

2 Cardinality

Now, I will talk about cardinality. This is really fun stuff, but I will not go through all the fun results here, only the results we need. The motivation of this is to show that there are, in a precise sense, “more” real numbers than natural numbers, so I can show that you cannot take a sum over all the real numbers of positive numbers and still get a finite number.

We consider two sets to be the same size if there is a one to one correspondence between their elements. This sounds obvious, but it shows that there are equally as many even positive integers as positive integers, because we can map 1—2 and 2—4 and 3—6 and so on, giving us a one to one correspondence.

Theorem. We cannot find a one to one correspondence between real numbers on any non-degenerate interval and positive integers.

The proof of this theorem is very famous in “popular math” and the slogan is “there are different sizes of infinity”.

Proof. We will do this for the interval (0,1) then note that we can get to any interval by scaling, or the entre reals by doing $-\cot(\pi x)$ for example. The property of there being a correspondence to positive integers is called being countable. But, I think listable is a better term.

So, suppose we do, in fact, have a list of all the real numbers between 0 and 1.

1. 0.**3**141592653589793...
2. 0.2**7**18281828459045...
3. 0.14**1**4213562373095...
4. 0.349**8**579345858968...
5. 0.9988**2**37478199283...

Now construct a new number where the n’th decimal digit after the point is not the red number, and such that we do not have infinite trailing 9’s or 0’s (to ensure the number has a unique decimal representation). For the numbers in our list we pick the one with trailing 0’s and not trailing 9’s whenever we have to make this choice). There are always many ways to do this. For example, adding (or subtracting if adding would get us a 9 or a 0) 1 to each bold digit we could have 0.48273... But this is clearly not on the list – It differs from everything on the list by at least one digit. Contradiction. Essentially, this makes it so if you think you found a way to list them, I can prove you are wrong.

□

We want to prove that you can’t add an unlistable amount of real numbers greater than 0 and get a finite result.

Now if we had a sum over the real numbers of positive numbers x, split them into subsets:

$$x \geq 1, \frac{1}{2} \leq x < 1, \frac{1}{3} \leq x < \frac{1}{2}, \frac{1}{4} \leq x < \frac{1}{3}, ...$$

Then one of these subsets has to have infinitely many elements. Why? If they all had finitely many elements then they would form a subset of this infinite table:

	1	2	3	4	5	6	7	8	9	...
Set 1										
Set 2										
Set 3										
Set 4										
Set 5										
Set 6										
Set 7										
Set 8										
...										

So I could then go along the table in a zigzag pattern like this in the order shown below, adding to my list whenever the elements are in the table, and not adding them otherwise.

	1	2	3	4	5	6	7	8	9	...
Set 1	1	2	6	7	15	...				
Set 2	3	5	8	14						
Set 3	4	9	13							
Set 4	10	12								
Set 5	11									
Set 6										
Set 7										
Set 8										
...										

All cells get reached eventually, so this would imply we can list the cells. But we are assuming we are adding one term for each real number, and we cannot list the real numbers. Therefore, one of the sets

$$x \geq 1,\ \frac{1}{2} \leq x < 1,\ \frac{1}{3} \leq x < \frac{1}{2},\ \frac{1}{4} \leq x < \frac{1}{3}.$$

has infinitely many elements, so the sum is infinite.

Now here is the key point: The set of all integers is countable by the list $0, 1, -1, 2, -2, 3, -3, ...$ so the set of pairs of integers is countable. Each pair (except those that contain a 0 in the second position) corresponds to a rational number and all rational numbers are gotten this way, in fact infinitely many times. Therefore, the set of rational numbers is also countable. This is important because it means if we have an increasing function then it has at most countably many discontinuities, as each discontinuity goes "through" a rational number. We will use this fact a lot when proving the central limit theorem.

3 Properties of integrals

3.1 Integrals and limits

Definition. Let f be any bounded function on $[a, b]$, then $\underline{\int_a^b} f(t)dt$ is the supremum over all partitions of $[a, b]$ of the integrals of sums of rectangles that underestimate the function at each partition. This is called the lower integral. We similarly define the upper integral as $\overline{\int_a^b} f(t)dt$. f is said to be Riemann integrable if these integrals are equal. By convention we require the interval and f to both be bounded.

Definition. We say a bounded set is elementary if it is a finite union of bounded open, closed, or half-open intervals and points and we write $m(E)$ for the sum of the lengths of its disjoint interval pieces with points having length 0. Finite unions, intersections and differences of elementary sets are elementary.

For any bounded set A define $c(A) = \sup \{m(E) : E \subseteq A \text{ and E is elementary}\}$

This is not the measure of A in general, just a function c that happens to coincide with the measure in some cases, but we do not rigorously define the measure here - we bypass that.

Definition. If B is a subset of A then $A \setminus B$ is everything in A but not in B.

Lemma. Let $A_1 \supseteq A_2 \supseteq ...$ be bounded sets with empty intersection, then $c(A_n) \to 0$

Proof. Suppose instead that $c(A_n) \geq \delta > 0$ for every n (this is what happens if it does not tend to 0 since it is a non-increasing sequence). Then for each n we can find a closed, bounded elementary set E_n inside A_n such that $m(E_n) > c(A_n) - \frac{\delta}{2^{n+1}}$ (if we could find ones with open or half-open intervals we could just chop a bit off the end). Put $H_n = E_1 \cap ... \cap E_n$. If J is an elementary subset of $A_i \setminus E_i$ then J and E_i are disjoint elementary subsets of A_i so

$$m(J) + m(E_i) = m(J \cup E_i) \leq c(A_i)$$

and therefore $m(J) < \frac{\delta}{2^{i+1}}$ by definition of $c(A_i)$ and E_i.

If H_n were empty then every elementary K inside A_n would satisfy $K = \bigcup_{i=1}^{n} (K \setminus E_i)$

Because $A_n \subseteq A_i$ for $i \leq n$ we have $m(K) < \sum_{i=1}^{n} \frac{\delta}{2^{i+1}} = \frac{\delta}{2}$ since K is an elementary set in every A_i. This contradicts $c(A_n) \geq \delta$ and therefore every H_n is non-empty and closed and so by the earlier lemma their intersection is non-empty which exhibits a point in all A_n which is a contradiction.

□

The following theorem is completely overpowered for making the applied nonsense in A levels/university Maths rigorous. I'll often refer to this theorem when I use it as the Dominated Convergence Theorem, but note that the actual statement about that theorem is for Lebesgue integrals, not Riemann integrals, so this isn't quite the dominated convergence theorem, and to prove it properly we would need to make this document about 3 times the length, not because it is hard but because there would be a lot of definitions we need and tedious checks that those definitions are sensible. We're proving this version to justify tools we use in A level maths such as the central limit theorem without that meaning measure theory.

Theorem. Suppose we have a sequence of integrable functions $f_n(x)$ which for all x converge to an integrable function $f(x)$ as n goes to infinity, and there is a function $g(x)$ which has a finite integral on the interval we are working in, and that $|f_n(x)| \leq g(x)$ for all n and all x, then we have that $\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n$, essentially giving us a condition for when we can swap limits and integrals.

This works for improper integrals over the real line $\mathbb{R}$ as long as the integral of the absolute value is finite (this is a condition of the theorem anyway)

Proof. **Step 1.** Start by assuming that the interval and dominating function is bounded. We will extend to a more general case later. Let $M > 0$ be a bound for all $|f_n|$ and thus for $|f|$ and let $[a, b]$ be the interval with $a < b$, and assume $f_n \to f$. Set $h_n = |f_n - f|$, then by the triangle inequality $0 \leq h_n \leq 2M$ and each h_n is Riemann integrable and $h_n \to 0$. We will prove that $\int h_n \to 0$.

Fix $\varepsilon > 0$ and set $\eta = \frac{\varepsilon}{2(b-a)}$. Define

$$A_n = \{x \in [a, b] : h_j(x) \geq \eta \text{ for at least one } j \geq n\}$$

By pointwise convergence, these sets are a decreasing sequence with empty intersection so we can pick N by the lemma such that if $n \geq N$ then $c(A_n) < \frac{\varepsilon}{4M}$.

Let $n \geq N$. Because the Riemann integral of non-negative h_n equals its lower integral, it is enough to bound the integral of every step function s with $0 \leq s \leq h_n$. Let $E = \{x : s(x) \geq \eta\}$, then E is elementary and $E \subseteq A_n$. Therefore,

$$\int_a^b s dx \leq 2Mm(E) + \eta(b-a) < \frac{2M\varepsilon}{4M} + \frac{\varepsilon}{2} = \varepsilon$$

Since this is true for every such function s, taking the supremum of the integral over all s the result follows.

Step 2. We want to extend to integrals over $\mathbb{R}$ with unbounded dominating functions.

The scope of our improper integrals will be that there are finitely many places where the dominating function blows up to infinity. Deleting a small neighbourhood about all such points makes an unbounded function bounded.

Fix some large integer M. The idea is as follows: Let $g \geq 0$ and let g be improperly integrable with a finite integral over $\mathbb{R}$. Then if g has some places where it blows up and the integral becomes improper then we need to extend it and modify f_n, f to be, say, 0 where it is undefined and then we will be able to take an interval or (later) a rectangle or cube with side length $\frac{1}{M}$ around it that we exclude from the domain. We will also make our domain be contained in $[-M, M]$ or the cube about the origin with side length $2M$. We will fix $\varepsilon > 0$. Once we have this domain restriction depending on M, we note the following 2 facts:

1. As $n \to \infty$ the integral of g, and thus f_n and f outside this restricted domain tends to 0 since g is improperly integrable. This must be the case since the improper integral is defined using a limit and the limit is assumed to exist. Thus, for some M sufficiently large, the restriction of g to the domain in question, g_M, satisfies $\int |g_M - g| < \varepsilon$.

2. The theorem holds on this domain since by step 1 it is a bounded interval with a bounded dominating function g.

Therefore the result follows

□

Now, if $f_n(x)$ is defined as n for 0<x<1/n then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral. It turns out that in this case we cannot find a function g such that for all n we have $f_n(x) \leq g(x)$ for all x but yet $\left|\int_0^1 g(x)\right| < \infty$, since $g(x)$ would have to behave like $\frac{1}{x}$ near 0. This is a standard textbook counterexample.

3.2 Multiple integrals

We will prove a special case of Fubini's theorem. The theorem says that if I have $\iint f(x, y)\, dxdy$ on a bounded rectangle then I can swap the integrals provided f is continuous. Specifically, $\int_{y_0}^{y_1} \left(\int_{x_0}^{x_1} f(x, y)\, dx\right) dy = \int_{x_0}^{x_1} \left(\int_{y_0}^{y_1} f(x, y)\, dy\right) dx$

Note that the same argument below can easily be adapted to work for swapping more integrals than 2.

Proof. Since f is continuous, given some $\varepsilon > 0$ there is a $\delta > 0$ such that $|\,(a_1, b_1) - (a_2, b_2)\,| < \delta$ implies $|f(a_1, b_1) - f(a_2, b_2)| < \frac{\varepsilon}{(1+x_1-x_0)(1+y_1-y_0)}$. The fact that there is a δ that works at every point follows from a result we will prove in chapter 5, and the fact that we are integrating on a closed bounded rectangle. Given this, make a partition of (x_0, x_1) by $(x_0, a_1, a_2, ..., a_{n-1}, x_1)$ such that each interval in the partition has length $< \frac{\delta}{\sqrt{2}}$so that any points in each rectangle are within δ of eachother and do the same for (y_0, y_1) where we name the points in the partition $(y_0, b_1, b_2, ..., b_{m-1}, y_1)$.

For convenience set $a_0 := x_0$, $a_n := x_1$, $b_0 := y_0$, $b_m := y_1$.

We now consider

$$\sum_{i=1}^{n} (a_i - a_{i-1}) \sum_{j=1}^{m} (b_j - b_{j-1}) \inf_{(x,y) \in [a_{i-1}, a_i] \times [b_{j-1}, b_j]} f(x, y)$$

and

$$\sum_{i=1}^{n} (a_i - a_{i-1}) \sum_{j=1}^{m} (b_j - b_{j-1}) \sup_{(x,y) \in [a_{i-1}, a_i] \times [b_{j-1}, b_j]} f(x, y)$$

Then these are within ε of eachother by construction

But then we note that sandwiched between them are the lower and upper integrals of $\int_{y_0}^{y_1} f(x, y)\, dy$ with respect to x and of $\int_{x_0}^{x_1} f(x, y)\, dx$ with respect to y. These integrals are continuous functions as changing the variable they depend on by $< \delta$ will change the integral by $< \varepsilon$ and there is a δ that works for any ε.

Therefore these lower and upper integrals will be forced to equal the iterated integral $\iint f(x, y)\, dxdy$ since they are sandwiched between things that get arbitrarily close to it.

□

Definition. Let g be a function, then we try to define the Riemann Stieltjes integral of $\int f(x)\, dg(x)$ as instead of taking $x_i - x_{i-1}$ we take $g(x_i) - g(x_{i-1})$. We will check shortly that the limits of integration work in the case we actually need. When we integrate with respect to a probability measure in Level 6.3, we are essentially setting g to the cumulative distribution function and redefining the "length" for the theorems above that were built on lengths of intervals - in particular it is useful to think of the length as the probability that a random variable is in said set, in which case ending the integral with $d\mu$ is the convention we will use.

We say that the limit exists if for every $a < b$ we choose a $\xi_{(a,b)}$ with $a \leq \xi_{(a,b)} \leq b$ and if we have a sequence of partitions P_n each with m_n intervals then the sum $S(n) = \sum_{i=1}^{m_n} f\left(\xi_{(x_{i-1},x_i)}\right)(x_{i-1} - x_i)$ approaches the same limit if $\sup\{x_i - x_{i-1}\} \to 0$ (i.e. the mesh of the partitions tend to 0) for your sequence of partitions, regardless of your choice of $\xi_{(a,b)}$

Proposition. If g is a cumulative distribution function and f is piecewise continuous with f, g never discontinuous at the same point then f is integrable on any bounded interval. We will call this integrating f with respect to a probability measure and put $d\mu$ at the end when we do this.

Proof. We can isolate the discontinuities of f, g by arbitrarily small amounts. The same as the level 4 proof that continuous functions are integrable relied on the fact that showing if our intervals get small enough f will differ by at most εon those intervals. We assumed without loss of generality that the integrals are from 0 to 1 and concluded from there, and the proof is exactly the same since the total change of g is never more than 1 as it is a cumulative probability distribution.

$\square$

Note that for improper integrals, provided the integral of the absolute value is finite, then if we make a big enough rectangle the integral of the absolute value of the outside things will go to 0, and the limit will not depend on what sequence of rectangles we take as long as they are increasing and eventually exhaust all points in $\mathbb{R}^n$. Therefore for improper integrals we will check for absolute convergence. We also need absolute convergence of the dy and dx integrals so that we can do the iterated integrals. For example, if we try to integrate the function that is only 1 when $y = 5$ and 0 everywhere else, then if we do the dy integral first we will be fine and if we do the dx integral first it will be undefined when $y = 5$. Concretely, we need $\int_{\mathbb{R}} \left|\int_{\mathbb{R}} |f(x,y)|\, dx\right| dy < \infty$ and $\int_{\mathbb{R}} \left|\int_{\mathbb{R}} |f(x,y)|\, dy\right| dx < \infty$ and $\int_{\mathbb{R}} |f(x,y)|\, dy < \infty$ for every x and $\int_{\mathbb{R}} |f(x,y)|\, dx < \infty$ for every y for the iterated integrals to work. Once we have that we may take limits.

Theorem. The above limit theorems work in $\mathbb{R}^n$ as well.

Sketch of proof. The lemma about closed sets holds with the same proof, defining elementary sets as finite unions of closed rectangles of any dimension $\leq n$ or points the lemma holds with an analogous proof.

Theorem. The above limit theorems work for Riemann-Stieltjes integrals with respect to probability measures, or for iterated integrals some of which are with respect to probability measures

Sketch of proof. The lemma about the sequence of sets holds with the same proof, only this time we define the length of elementary sets by their cumulative distribution function weight. The rest of the proof then proceeds the same way. The extension to improper integrals proceeds the same way.

Theorem. The above "integral swap" theorem works if one of the integrals is with respect to a probability measure over $\mathbb{R}$ and we are integrating a function with no discontinuities in common with points that are discontinuities of the cdf.

Sketch of proof. The same as the original proof with $b_i - b_{i-1}$ replaced by $P(b_i \geq X > b_{i-1})$ where X is the random variable corresponding to the probability measure. We also need the fact that if $R \to \infty$ then $P(|X| \geq R) \to 0$, as this allows passing the limit to infinity. The sentence "These integrals are continuous functions as changing the variable they depend on by $< \delta$ will change the integral by $< \varepsilon$ and there is a δ that works for any ε" from the original proof still works because the total variation of probability is 1 so the discrepency from the upper and lower sums, once you add up the contributions for the entire partition, is still $< \varepsilon$.

4 Intuition for complex integration

Also, in my exponentials and logarithms video, I briefly mentioned that if a function has a single-valued antiderivative then the integral along a path of that function does not depend on the path. By "integrate along a path", I mean take the sum of the function times the complex displacement you move by on the path (eg if you move from $1.07+3.12i$ to $1.08+3.14i$ you add $(0.01+0.02i)f(1.07+3.12i)$), then take the limit of these sums as these distances go to 0. Intuitively, similar to actual integration, taking the sum like this moves you along the antiderivative, which if it is single valued (unlike log, for example) will not allow you to possibly have path dependence.

Note: We will formalize this in level 8.3, the complex analysis tripos course. For now this tells you why it is true.

5 Uniform continuity

Definition. A function is said to be uniformly continuous if given an $\varepsilon > 0$ in the definition of continuity not only are we able to pick a δ for each x, but a δ that works for all x. An example of a function that is not uniformly continuous is 1/x on the open interval (0,1), since if I pick a certain ε, then no matter how small I pick δ, I can go close enough to 0 that the $|x-a|<\delta$ implies $|f(a)-f(x)|<\varepsilon$ condition is not satisfied, so I cannot pick a δ that works for all x.

Specifically,

Continuous: A function is continuous if for all x in its domain it is true that for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-y| < \delta \Longrightarrow |f(x) - f(y)| < \varepsilon$

Uniformly continuous: A function is uniformly continuous if for all $\varepsilon > 0$ there is a $\delta > 0$ such that for all x in its domain it is true that$|x-y| < \delta \Longrightarrow |f(x) - f(y)| < \varepsilon$

This difference is subtle but it is important. For example, $\frac{1}{x}$ for $x > 0$ is continuous but not uniformly continuous, since for each ε if I pick a δ and go close enough to 0 it will stop being a δ that works. However, there is something useful we can say on the matter.

Theorem. Suppose that f is a function which is continuous on a closed bounded interval, then it is uniformly continuous

Proof. Since we are assuming our function is continuous on a closed bounded interval, the Bolzano Weierstrass theorem applies by boundedness, and the subsequence in question converges to a limit in the interval by closure of the interval.

We will see similarities to the level 4 proof that continuous functions are integrable.

Assume for a contradiction that our function is continuous but not uniformly continuous on our closed bounded interval. Then there is an $\varepsilon > 0$ such that if I pick $\delta = \frac{1}{n}$ there is some x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| > \varepsilon$. We know that x_n and y_n both have convergent subsequences converging to places on our closed interval, and in fact they converge to the same place since the difference between x_n and y_n gets arbitrarily small. If the subsequences in question are x_{n_i} and y_{n_i} and their limits are x, then $f(x_{n_i}) \to f(x)$ and $f(y_{n_i}) \to f(x)$ because continuous functions preserve limits. But $f(x_{n_i})$ and $f(y_{n_i})$ somehow converge to the same limit but are always $> \varepsilon$, which is a contradiction.

□

6 Calculus results

6.1 Big O and little o notation

Definition. A function $g(x)$ is said to be $O(f(x))$ if when x is sufficiently close to x_0, $|g(x)|$ is bounded by $M|f(x)|$ where M is some fixed positive constant. If $f(x)$ is not 0 in the vicinity of x_0 we can write that $|\frac{g(x)}{f(x)}| < M$ when x is sufficiently close to x_0. We can also say, and this is often done in computer science when analyzing how long an algorithm will take, that a function $g(x)$ is $O(f(x))$ as x goes to infinite if when x is large enough, $|g(x)|$ is bounded by $M|f(x)|$. The value of x_0 in a specific situation is often implied by context. Sometimes people write $g(x) = O(f(x))$

Definition. Little o notation is like big O notation except if $g(x)$ is $o(f(x))$ at x_0 it means that $g(x)$ gets much smaller than $f(x)$ when x is sufficiently close to x_0. Precisely, it means that M can be made arbitrarily small by making x sufficiently close to x_0, or by making x sufficiently large in the case x_0 is infinity.

Note that if $f(x) = o(g(x))$, then we necessarily also have $f(x) = O(g(x))$.

Example. $x \neq O\left(x^2\right)$ as $x \to 0$ since x^2 is much smaller than x. In fact, $x^2 = o\left(x\right)$ as $x \to 0$.

$x = O\left(x^2\right) = o\left(x^2\right)$ as $x \to \infty$ but $x^2 \neq O\left(x\right)$ as $x \to \infty$.

$x = O\left(\sqrt{x}\right)$ as $x \to 0$.

$\sin(2x) = O\left(x\right)$ as $x \to 0$, but $\sin(2x) \neq o\left(x\right)$ as $x \to 0$.

$x \neq O\left(x\sin(x)\right)$ as $x \to \infty$, because although x does not seem to be much larger than $x\sin(x)$, the ratio $\frac{x}{x\sin(x)}$ is unbounded when x gets close to integer multiples of π.

Also, a principle here is the idea that a polynomial is dominated by its leading term. For example

$2x^3 + 4x + 12 = O(x^3)$ as $x \to \infty$ because clearly eventually x^3 exceeds $4x + 12$ so the whole thing is less than $3x^3$ so we can put $M = 3$. Also, big o notation can capture the idea that exponentials beat polynomials (by the way you can prove this as $\frac{a^x}{x^b}$ goes to infinity as x goes to infinity which you can see if you apply l'hopital's rule at least b times), as we can write that $P(x) = o(\exp(x))$ for any polynomial P. Also, $\log\left(x\right) = o(x^{\varepsilon})$ for all positive ε and this can be proven the same way, or by thinking of it intuitively as a sort of inverse of the exponentials beat polynomials idea. Note $\exp(x)$ means e^x.

Another idea is that non-zero constants don't matter here since we're looking at the bigger picture with the growth rate of these functions.

Example. Since $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$, $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} - f'(x) = 0$ so $\frac{f(x+h)-f(x)}{h} - f'\left(x\right) = o(1)$ as $h \to 0$ by definition. Therefore, $f\left(x+h\right) - f(x) - hf'\left(x\right) = o(h)$ by multiplying by h on both sides, so we can write that $f\left(x+h\right) - f\left(x\right) = hf'\left(x\right) + o(h)$, which looks a lot like what I did when I justified differentiating infinite power series.

6.2 Taylor's theorem

Definition. A taylor polynomial of degree n is a taylor series for a function that is n times differentiable but only up to the term in x^n. We write this as $T_n(f)(x)$. This is a polynomial which gives an approximation of f near some point x_0, with error equal to $o((x-x_0)^n)$.

Claim. If f is n times differentiable at x_0 then $f\left(x\right) - T_n\left(f\right)\left(x\right) = o((x-x_0)^n)$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If f is n+1 times continuously differentiable on the interval $[x_0,\ x]$ then $f\left(x\right) - T_n\left(f\right)\left(x\right) = O((x-x_0)^{n+1})$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If the conditions for claim 2 hold, then there exists some $t \in (x_0,\ x)$ such that $f\left(x\right) - T_n\left(f\right)\left(x\right) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(t)$.

We will prove this.

Lemma. (Extreme value theorem) Any function continuous on a finite closed interval is bounded and actually attains its minimum and maximum.

Proof. Note: It is very important that the interval is closed. On the open interval (0, 1) we can define 1/x which is continuous on that interval, but not on the closed interval [0,1] because the closed interval contains $x = 0$ but the open interval does not. But there's not much you can do to make a continuous unbounded function defined on a finite closed interval – Try it!

We will use the fact that every bounded sequence has a convergent subsequence since that was proven in level 4.

Suppose for a contradiction f is unbounded. Then for all n, we can choose x_n in [a,b] with $|f(x_n)| > n$. Now pick a convergent subsequence x_{n_k} of the sequence formed by each x_n, and call its limit L, which is in $[a, b]$ since $[a, b]$ is closed so limits of sequences in [a,b] are in [a,b]. By continuity, $f(x_{n_k}) \to f(L)$, but since $f(x_{n_k})$ is unbounded, it follows that this is a contradiction, since $f(L)$ cannot be defined.

Important note: Bolzano weierstrass holds for bounded sequences of points in k dimensions, and if the bounded set in question is closed then the limit of the subsequence will also be in the set – the reason is that there is a subsequence whose x coordinate converges and a subsequence of that whose y coordinate converges etc, and thus the above theorem holds for functions of more than one variable, as well as the theorem about uniform continuity and about Riemann integrals for continuous functions, as the definition of continuity is analogous. This is important since we use these theorems in the context of functions of multiple variables.

We will now prove that the maximum is attained. By boundedness there is a least upper bound M. Therefore there is a sequence of increasing points $M_1, M_2, \ldots$ converging to M that are values of the function. For each of these points pick an input value t_k of f such that $f(t_k) = M_k$. Now t_k has a convergent subsequence converging to t and by continuity $f(t) = M$, meaning the maximum value of a function on a closed interval is attained. It is by closedness that we can guarantee t is in our interval.

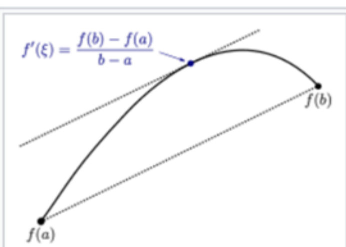
$\square$

Lemma. (Mean value theorem)

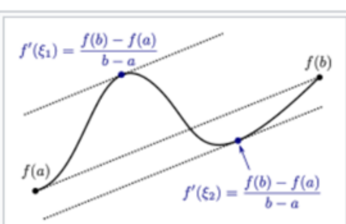
If we have a continuous function on [a,b] differentiable on the open interval (a,b) then at some point in (a,b) the derivative is equal to its mean value - specifically the slope of the line joining the endpoints.

This lemma assumes the values are real, so everything proven here applies to real numbers, but not general complex numbers.

Proof. First of all, a picture (Figure 1, from Wikipedia):



The function f attains the slope of the secant between a and b as the derivative at the point $\xi \in (a, b)$.



It is also possible that there are multiple tangents parallel to the secant.

Figure 1

You can see that it's kind of obvious that the function's derivative attains its mean value, since the function is differentiable everywhere by assumption so there are no spikes. This is essentially what the theorem says. However, it is not completely obvious as derivatives can be discontinuous.

To prove this, we want to subtract the line from $(a, f(a))$ to $(b, f(b))$ from y. This line has the slope we want so we want to prove that our new function, which is 0 at a and b, attains a derivative of 0 between a and b. This new function is either constant, in which case the theorem is obvious, or it goes above 0 at some point (If not prove the theorem for minus the function). Since it is continuous on a closed, bounded interval, it is bounded, and has either a maximum or a minimum. At such a point, the derivative exists by our hypothesis, and it can't possibly be something other than 0. Why? Suppose the derivative were positive at a maximum point c, then by definition of limits, $\frac{f(x)-f(c)}{x-c}$ would be positive if we make x close enough to c with $x>c$, implying $f(x)>f(c)$ which is a contradiction.

In particular, we have Rolle's theorem, which says that if $f(a) = f(b) = 0$ and f is continuous on $[a, b]$ and differentiable on (a, b) then there is a point $a<x<b$ such that $f'(x) = 0$

□

Lemma. Let f be continuous on $[a, b]$ and n-1 times continuously differentiable, but also n times differentiable on the open interval (a,b), and suppose also that

$$f(a) = f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = f(b) = 0$$

then there is some x in (a,b) such that $f^{(n)}(x) = 0$.

Proof. Induct on n. The $n = 1$ base case is just Rolle's theorem. Suppose we have $k < n$ and $x_k \in (a, b)$ such that $f^{(k)}(x_k) = 0$. Since $f^{(k)}(a) = 0$, we can find $x_{k+1} \in (a, x_k)$ such that $f^{(k+1)}(x_{k+1}) = 0$ by Rolle's theorem. So the result follows by induction.

□

Example. The function $x^3 - x^2$ has its first derivative and value equal to 0 when $x = 0$, and its value equal to 0 at $x = 1$, so between 0 and 1 it must have a point of second derivative zero. Indeed it does when $x = \frac{1}{3}$.

Proposition. If $f(x)$ is n times differentiable at x_0 then

$$R(x) := f(x) - f(x_0) - (x - x_0) f'(x_0) - \frac{(x - x_0)^2}{2!} f''(x_0) - \cdots - \frac{(x - x_0)^n}{n!} f^{(n)}(x_0)$$

is $o((x - x_0)^n)$ as $x \to x_0$.

Proof. The first n derivatives of the above expression are all 0 when evaluated at x_0 because for k between 0 and n inclusive the $\frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$ term has k'th derivative equal to $f^{(k)}(x_0)$ and all its other derivatives are 0 at x_0 by repeated application of the power rule so therefore it cancels the k'th derivative of the $f(x)$ term to make it equal to 0.

So, by the definition of o, we need to show that $\frac{R(x)}{(x-x_0)^n} \to 0$ as $x \to x_0$. This is fine as $(x - x_0)^n$ is not 0 when you make it arbitrarily small so it can go in the denominator.

Now, if we can show that any function whose first n derivatives are all 0 at x_0 is $o((x - x_0)^n)$, we will be done. It is clear that a function whose value and first derivative is 0 at x_0 is $o(x - x_0)$ since $\frac{g(x)}{x-x_0} = \frac{g(x)-g(x_0)}{x-x_0} \to g'(x_0) = 0$, so this theorem is true if n=1. But now suppose that we know that this theorem is true for n=k, then we will prove it is also true for n=k+1, therefore proving this by induction. Let g(x) be such that at x_0 its value and first k+1 derivatives are all 0. By the induction hypothesis, g'(x) has its first k derivatives and value equal to 0 at x_0 so $g'(x_0 + h) = h^k \varepsilon(h)$ with $\varepsilon(h) \to 0$ as $h \to 0$, by the definition of little o notation and the induction hypothesis.

By the mean value theorem for some θ between 0 and 1 we have $hg'(x_0 + \theta h) = g(x_0 + h) - g(x_0)$ and $g'(x_0 + \theta h) = (\theta h)^k \varepsilon(h)$, meaning that by the induction hypothesis

$$g(x_0 + h) - g(x_0) = h\left((\theta h)^k \varepsilon(\theta h)\right) = \theta^k h^{k+1} \varepsilon(\theta h) = o(h^{k+1})$$

So induction done and proof complete.

□

Remark. Just because each remainder is small does not mean that the taylor series converges, since convergence at some x requires that there, the remainder is uniformly small, but it could be that the interval where the remainder is small shrinks and is zero in the limit. There are functions such that the taylor series does not converge in any interval despite the functions being infinitely differentiable (these are difficult to construct explcitly but the point is it is not automatic that a taylor series that is convergent exists), or functions where the taylor series converges but does not equal the function, like $\exp(-x^{-2})$ around x=0. However, it turns out that if the function has a complex derivative the taylor series always converges in some interval, see complex analysis in a later level. I'm not proving it because we don't need to use it for this purpose, but it trivializes all of these results.

Lemma. Let f be n+1 times differentiable on [a,b] with it's (n+1)'th derivative continuous. Then we have that

$$f(b) = f(a) + (b-a) f'(a) + \frac{(b-a)^2}{2!} f^{(2)}(a) + \cdots + \frac{(b-a)^n}{n!} f^{(n)}(a) + \int_a^b \frac{(b-t)^n}{n!} f^{(n+1)}(t)\, dt$$

Proof. We will do this by induction on n. When n=0 the theorem says

$$f(b) = f(a) + \int_a^b f'(t)\, dt$$

Which is known to be true. Now we will do integration by parts to get

$$\int_a^b \frac{(b-t)^n}{n!} f^{(n+1)}(t) dt = \left[\frac{-(b-t)^{n+1}}{(n+1)!} f^{(n+1)}(t) \right]_a^b + \int_a^b \frac{(b-t)^{n+1}}{(n+1)!} f^{(n+2)}(t) dt$$

$$= \frac{(b-a)^{n+1}}{(n+1)!} f^{(n+1)}(a) + \int_a^b \frac{(b-t)^{n+1}}{(n+1)!} f^{(n+2)}(t) dt$$

So the lemma follows by induction.

□

Proposition. If $f(x)$ is n+1 times continuously differentiable in an open interval containing x_0 then

$$f(x) - f(x_0) - (x - x_0) f'(x_0) - \frac{(x-x_0)^2}{2!} f''(x_0) - \cdots - \frac{(x-x_0)^n}{n!} f^{(n)}(x_0)$$

is $O((x-x_0)^{n+1})$ as $x \to x_0$.

Proof. Let $x_0 = a$ and let b be such that the derivative is continuous on $[a,b]$ and let $x \to a$. The lemma above immediately implies the result since $f^{(n+1)}$ is continuous on the closed interval $[x_0, x]$ and thus bounded there, say it is bounded in absolute value by M. Then the integral is at most $\int_a^x \frac{(x-t)^n}{n!} M dt = M \frac{(x-a)^{n+1}}{n+1!}$ which is $O((x-a)^{n+1})$ as required. As we

□

Proposition. $f(x) - f(x_0) - (x-x_0) f'(x_0) - \frac{(x-x_0)^2}{2!} f''(x_0) - \cdots - \frac{(x-x_0)^n}{n!} f^{(n)}(x_0) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(t)$ for some t(x) between x_0 and x, provided the first n+1 derivatives of f exist and are continuous from x_0 to x. (In fact we don't need the $n+1$'th derivative to be continuous, but merely for it to exist on the open interval (x_0, x) as that is enough to apply Rolle's theorem).

Proof. The n=0 case is just the mean value theorem.

Consider $g(x) := f(x) - f(x_0) - (x-x_0) f'(x_0) - \frac{(x-x_0)^2}{2!} f''(x_0) - \cdots - \frac{(x-x_0)^n}{n!} f^{(n)}(x_0)$. This has its value and first n derivatives at x_0 equal to 0 by the same reasoning as we gave in one of the earlier proofs

Set $\frac{g(x)}{(x-x_0)^{n+1}} = C$ when x is our constant. Then $g(x) - C(x-x_0)^{n+1}$ as a function of x (so x is no longer constant) is 0 at x from how we defined C and has its first n derivatives and value equal to 0 at x_0 meaning it satisfies the conditions for generalized rolle's theorem. Therefore, there exists a point y between x_0 and x such that the n+1'th derivative of

$g(y) - C(y-x_0)^{n+1}$ is 0. But the derivative of the second term is just C(n+1)! by the power rule, thus the n+1'th derivative of g at y is also C(n+1)!, thus $C = \frac{g^{(n+1)}(y)}{(n+1)!}$. Therefore, since we have that $g(x) - C(x-x_0)^{n+1}$ is 0 at x from earlier, we thus have that

$$g(x) = C(x-x_0)^{n+1} = \frac{g^{(n+1)}(y)}{(n+1)!}(x-x_0)^{n+1}$$

Which is exactly what we wanted to prove.

□

Example. There exists some point between 0 and 1 such that $\frac{e^y}{2} = e-2$, because if $x_1 = 1$ and x_0 =0, then $e^{x_1} - x_1 - 1 = \frac{x_1^2}{2}e^y$ for some y between 0 and 1, but x=1 so this simplifies to $\frac{e^y}{2} = e-2$.

6.3 L'hopital's rule

Lemma. (Cauchy mean value theorem) Let f, g be functions taking real values continuous on $[a,b]$ and differentiable on (a,b). Then there exists a c in (a,b) such that $g'(c)(f(b)-f(a)) = f'(c)(g(b)-g(a))$

Proof. Try the function $h(x) = [g(x) - g(a)][f(b) - f(a)] - [g(b) - g(a)][f(x) - f(a)]$.

Now $h'(x) = g'(x)(f(b) - f(a)) - f'(x)(g(b) - g(a))$

This is 0 at some point because $h(a) = h(b)$ and Rolle's theorem is true.

□

Proof of L'hopital's rule

Proof. Suppose that the following hold:

1. $f(a) = g(a) = 0$
2. $g(x)$ is not 0 if x is close to a
3. $\lim_{x\to a^+} \frac{f'(x)}{g'(x)}$ exists and is equal to l.
4. f and g are differentiable in some open interval $(a, a+\eta)$ and continuous at a.

By the Cauchy mean value theorem $g'(t)(f(b)) = f'(t)(g(b))$ for some t between a and b, for any $a+\eta > b>a$. Rearranging gives that

$$\frac{f(b)}{g(b)} = \frac{f'(t)}{g'(t)}$$

I emphasise that t depends on b but for each fixed b it exists and is between a and b. Therefore, as $b \to a$, so does t.

Thus, $\frac{f(b)}{g(b)} \to l$ as $b \to a$.

If instead the limit is from the left the proof is analogous.

□

7 Uniform convergence

We will define uniform convergence.

Here are the definitions.

Convergence: At every x, for every $\varepsilon > 0$ there exists an $N > 0$ such that $n > N$ implies $|f_n(x) - f(x)| < \varepsilon$

Uniform Convergence: F or every $\varepsilon > 0$ there exists an $N > 0$ such that for every x, $n > N$ implies $|f_n(x) - f(x)| < \varepsilon$

We know about how pointwise convergence works – a sequence of functions converges to a function if it converges at every value. We can say "For each point, there is an $\varepsilon > 0$ such that for all x, our sequence of functions $f(x)$ is eventually, after some n, within ε of the limit function". This is different from uniform convergence in the sense that uniform convergence requires n not to depend on x. We need the function to eventually get arbitrarily close at all points at once, not just at each point. An example of a sequence of functions that converges pointwise but fails this is the following:

$$f_n(x) = \begin{cases} 0 < x < \frac{1}{n} : 1 - nx \\ \text{Otherwise} : 0 \end{cases}$$

This looks like the functions in these images:

Figure 2 shows examples of this for n=1, 2, 10.

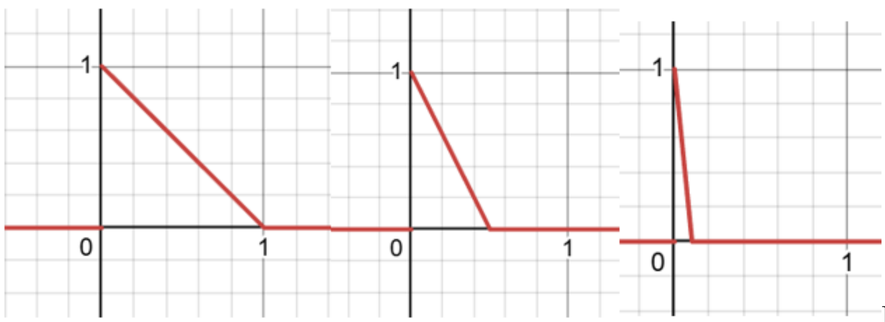


Figure 2

The point is: For any point you choose, say $k > 0$, once $n \geq \frac{1}{k}$, it will get to 0. So the functions converge pointwise to 0. It seems like we can pick x=0 and it will converge to 1 but we defined f(0) to be 0. We do not converge uniformly to 0 as we never satisfy that all points get arbitrarily close to 0.

It is useful because, for example, if we want to swap a limit with an integral we know that the total error will eventually get small, rather than just the error at each point.